MDS initialization and neighborhood sensitivity on five sampled saddle surfaces

Source: `~/current_projects/grip/papers/grip-software-paper/reproducibility/experiments/mds-initializer-sensitivity/report.tex`

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Question and main findings

Does the manuscript's edge-KK path-fidelity result depend on classical scaling as its initializer, and how sensitive is it to the symmetric k-nearest-neighbor graph? We compare classical MDS and metric stress MDS, each before and after edge-KK, on the same five existing samples. The generating coordinates provide a fifth control. These are finite, exploratory comparisons, not a theorem about all MDS or edge-KK solutions.

Across all 25 evaluated graphs, edge-KK reduces fixed-path error in 25/25 classical-initialized and 25/25 stress-initialized comparisons. It increases chord profiled Stress-1 in 25/25 and 25/25 comparisons, respectively. Stress initialization gives the lower final path error on 23/25 graphs (14/15 at the selected and nearby choices). This is an empirical preference, not uniform superiority over graph choices or global minimizers.

At the selected graphs, median path error is 3.451% for classical MDS and 2.264% for stress MDS; after edge-KK it is 0.245% and 0.164%, respectively. The corresponding median chord scores increase from 3.459% to 4.715% and from 2.803% to 5.213%.

Geometric recovery is less consistent. The stress-initialized edge-KK coordinate error at selected graphs ranges from 4.60–27.54% after similarity alignment. Good fixed-path agreement therefore does not establish correspondence or surface recovery. Across the selected and nearby choices, its largest coordinate error is 28.11%.

What was held fixed

Each sample contains 1,000 area-uniform points on $z = 0.8(x^2 - y^2)$ over $[-1, 1]^2$. Graphs use exact ambient nearest neighbors, symmetric union, ambient chord edge lengths, no pruning, and the original component-MST repair rule. The selected values are $71, 70, 67, 72, 73$. For each cloud, the frozen neighborhood choices are 32, $k^\star - 5$, $k^\star$, $k^\star + 5$, and 80. No embedding outcome was used to choose them. The saved 1%-of-minimum graph/reference-error plateaus are respectively 71–72, 64–71, 65–68, 70–73, and 65–73. The ± 5 choices probe more broadly than these narrow plateaus.

The MDS inputs here are graph shortest-path distances. The numerical smooth-surface geodesics are independent evaluation references. This matches the manuscript's $X \to G \to Z$ workflow. It is distinct from the future experiment that feeds smooth-geodesic distances directly into a graph construction and studies increasing paraboloid/saddle radius.

Methods and score definitions

Classical scaling uses `grip::classical.mds()`. Stress MDS uses `grip::metric.mds()`, with `smacof::mds(type="ratio")` as its majorization backend; ratio disparities preserve dissimilarity ratios, with internal target normalization [Mair et al., 2022]. The grip adapter restores the raw-stress-optimal coordinate scale in the original input units. Each graph receives a classical start and five Gaussian starts, each with a 1,000-iteration cap and tolerance 10^{-8}. The least achieved raw stress selects the result. Random configurations are paired across k within each cloud. Neither multiple starts nor the stopping tolerance establishes global optimality.

Both edge-KK branches use the original density-to-uniform continuation $(0, .25, .5, .75, 1)$, 200 steps per stage, profiled target scale, and zero edge-length stabilizer. At selected graphs, separate checks continue the selected MDS solution for 2,000 iterations at tolerance 10^{-10} and each edge-KK solution for 1,000 uniform-stiffness steps. They do not replace the primary fits or alter their initialization.

Write d for input target lengths and y for observed lengths. The fitted target multiplier is $a = y^\top d/(d^\top d)$. Edge and fixed-path errors are $\|y - ad\|/\|ad\|$, with separately fitted scales. Chord profiled Stress-1 is $\|y - ad\|/\|y\|$. Raw chord stress is $\|y - d\|^2$ at returned coordinate scale. Its target-normalized RMSE and literal unprofiled Stress-1 are also saved. With free coordinate scale, the minimum raw stress divided by $\|d\|^2$ equals squared profiled Stress-1; this identity is independently checked.

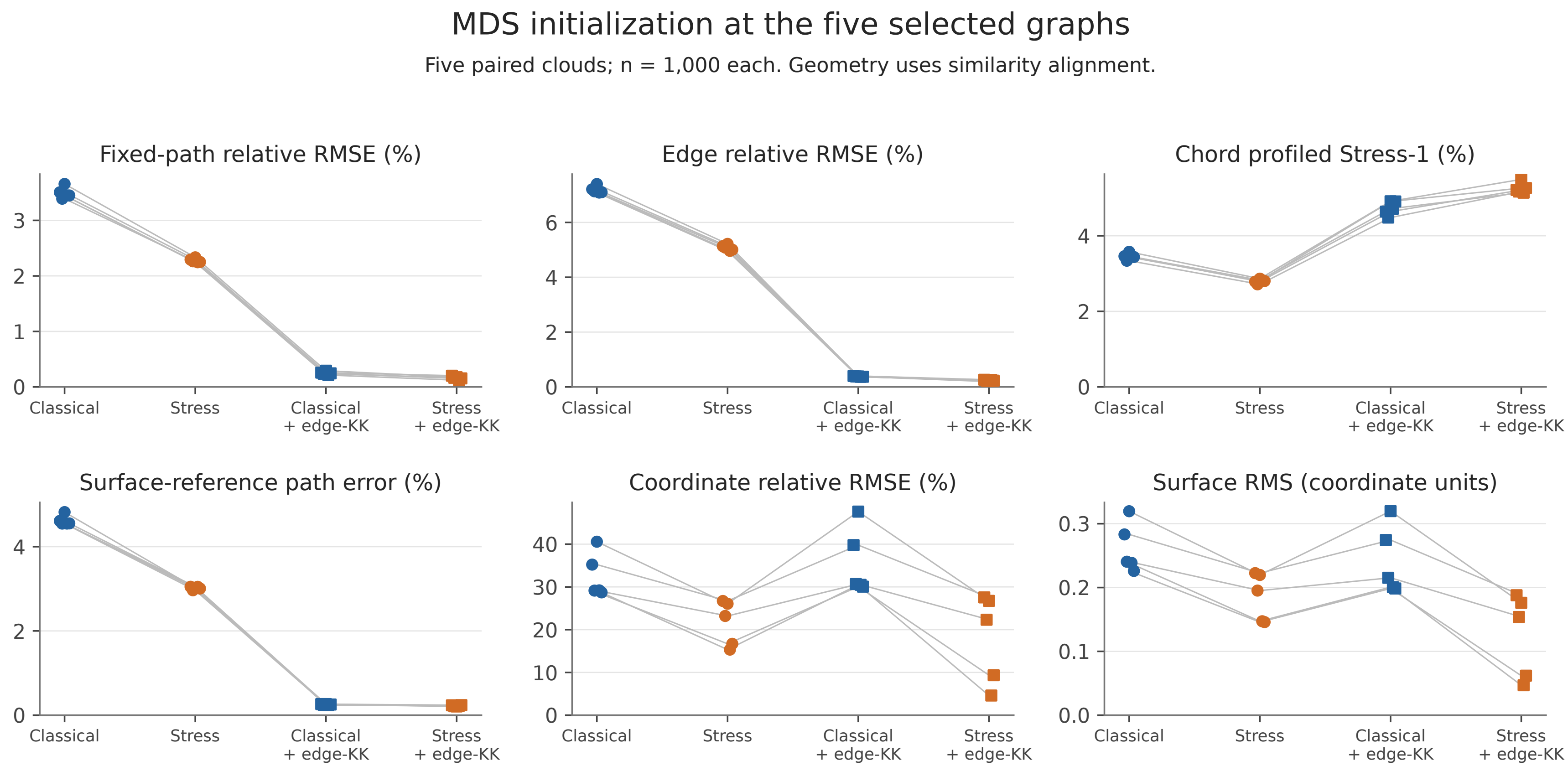
All 499,500 unordered pairs use a retained input-graph route, fixed across layouts on each graph. No embedded shortest paths are recomputed. For $X \to G$ error, strict graph distances are compared directly with the same 119,744 numerical surface-reference pairs per cloud. For $X \to Z$ path error, embedded path lengths are divided by the edge-fitted scale before that comparison; no scale is fitted to the surface reference.

Coordinate error uses known row correspondence and divides alignment RMSE by the original RMS radius. Rigid alignment permits rotation/reflection and translation; similarity alignment also permits uniform scaling. Surface RMS averages the two directional mean squared closest-triangle distances after coordinate alignment. Each embedded mesh uses original parameter-space Delaunay connectivity; the reference subdivides those triangles twice and lifts them onto the saddle. There are 8,000 area-uniform samples per direction. The original mesh is a nonzero discretization control. Surface Monte Carlo SE describes sampling error only; this statistic does not certify absence of folds.

Numerical input and validation scope

The retained-path preparation can contain tiny directional distance differences from its near-tie rule. A preliminary fit exposed an asymmetry of about 2.67×10^{-8}, which the MDS validator correctly rejected. All final fits therefore use the saved strict graph-distance matrices, symmetrized only at floating-point scale, for both MDS algorithms. Retained route lengths continue to define path diagnostics. The earlier partial run is excluded. No symmetry validation was relaxed. Historical classical distances and scores are checked to 10^{-7}; independent path sums and common package scores are checked more tightly. The saved numerical smooth-reference validation is reused and is not a certified all-pairs accuracy bound.

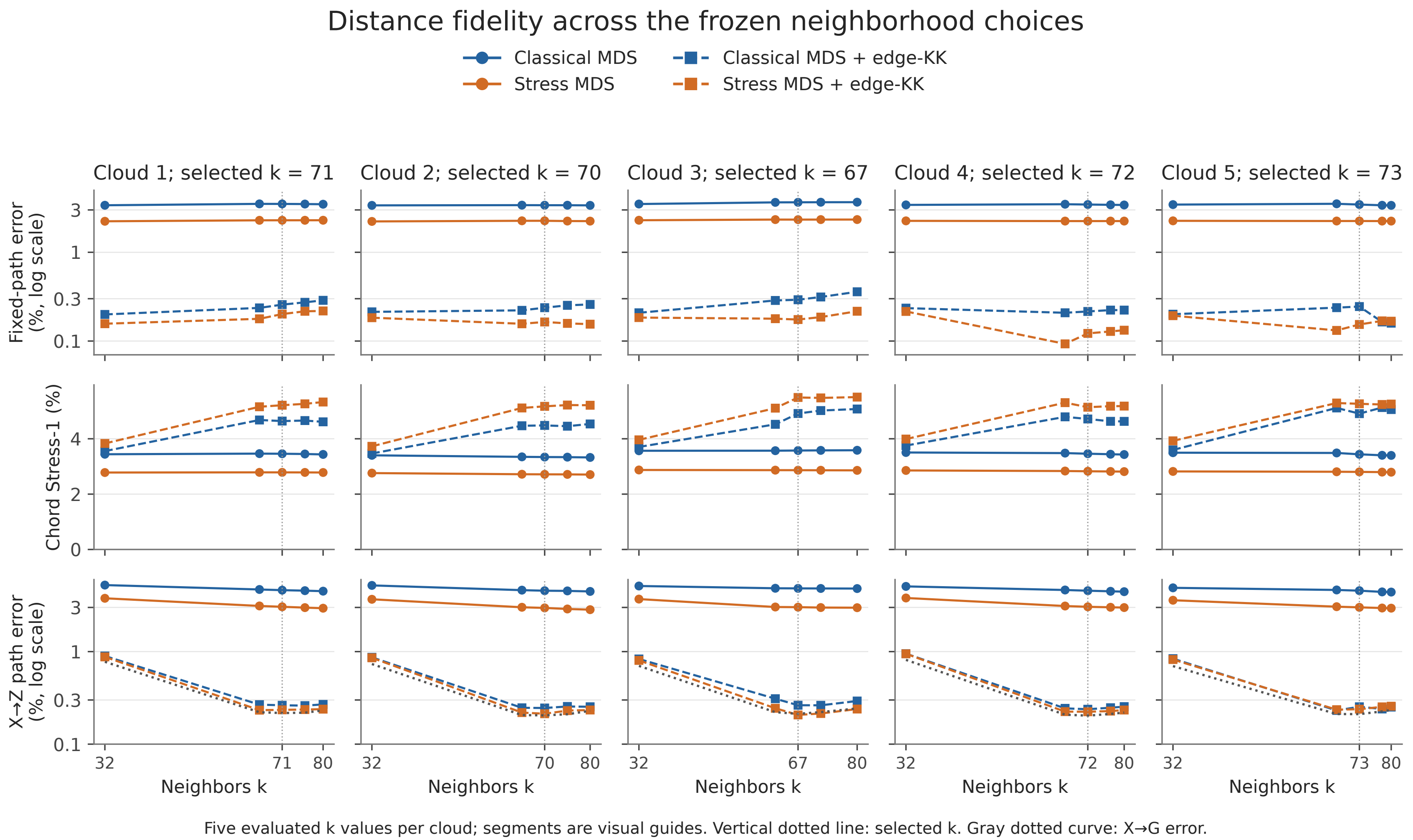
Selected-graph comparison



Five dots per method represent the five samples; gray lines connect the same sample. All distance scores are percentages except surface RMS, which is in coordinate units. Geometry uses similarity alignment. The original control has effectively zero edge/path/coordinate error; its surface discrepancy is nonzero because a piecewise-linear sample mesh approximates the curved reference. The table reports medians across the five clouds.

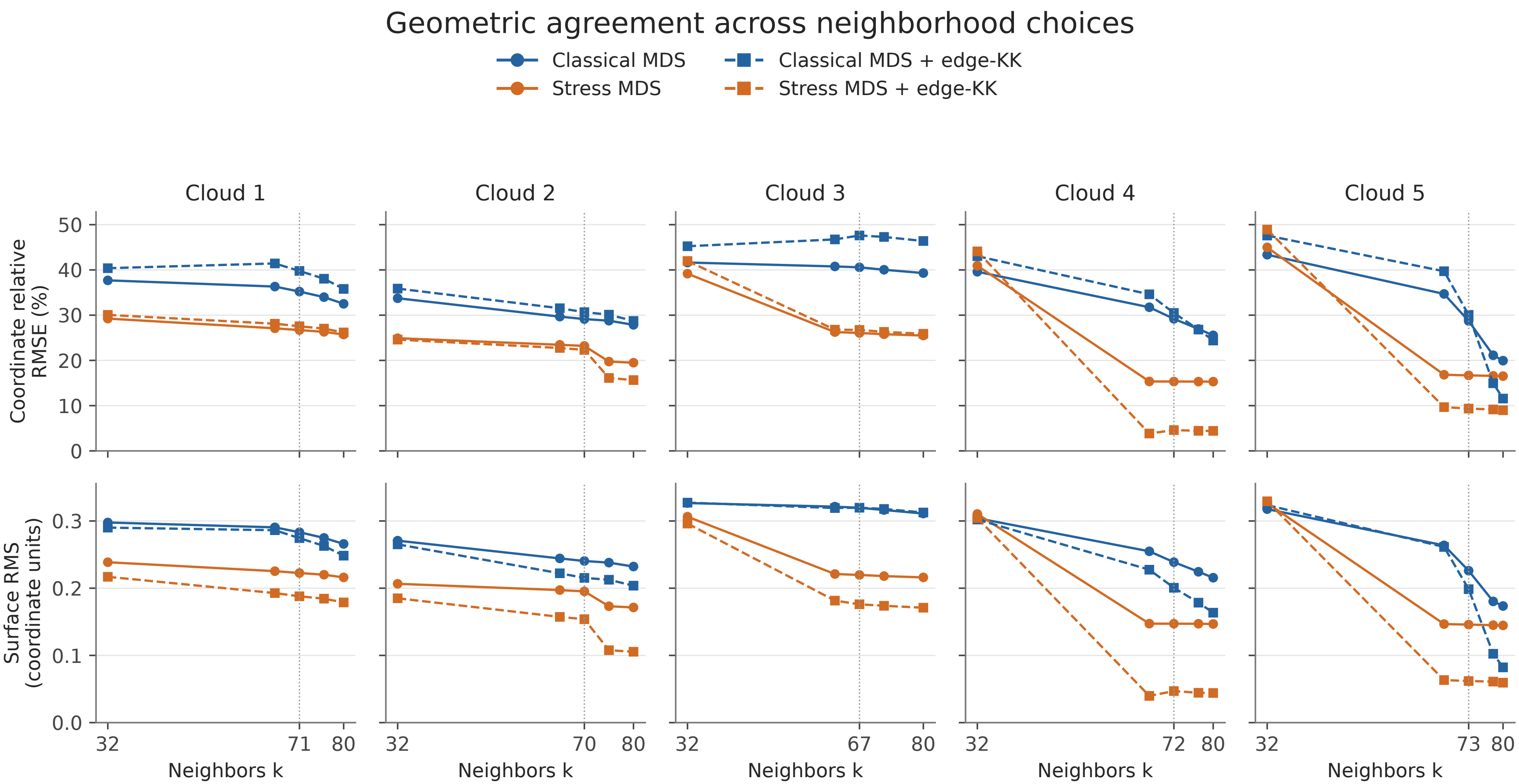
Method	Path (%)	Edge (%)	Chord (%)	$X \to Z$ (%)	Coord. (%)	Surface RMS
Classical MDS	3.451	7.133	3.459	4.553	29.190	0.240
Stress MDS	2.264	5.081	2.803	3.020	23.204	0.195
Classical + edge-KK	0.245	0.382	4.715	0.253	30.671	0.215
Stress + edge-KK	0.164	0.227	5.213	0.224	22.324	0.154

Sensitivity of distance fidelity to the graph



All four candidates are paired within each cloud and graph. Path-error axes use logarithmic scales. The vertical dotted line marks the originally selected k. The gray dotted curve in the bottom row is $X \to G$ error. Segments connect evaluated settings and do not claim behavior at every intermediate integer. Low graph-to-layout path error cannot remove error already introduced by graph construction.

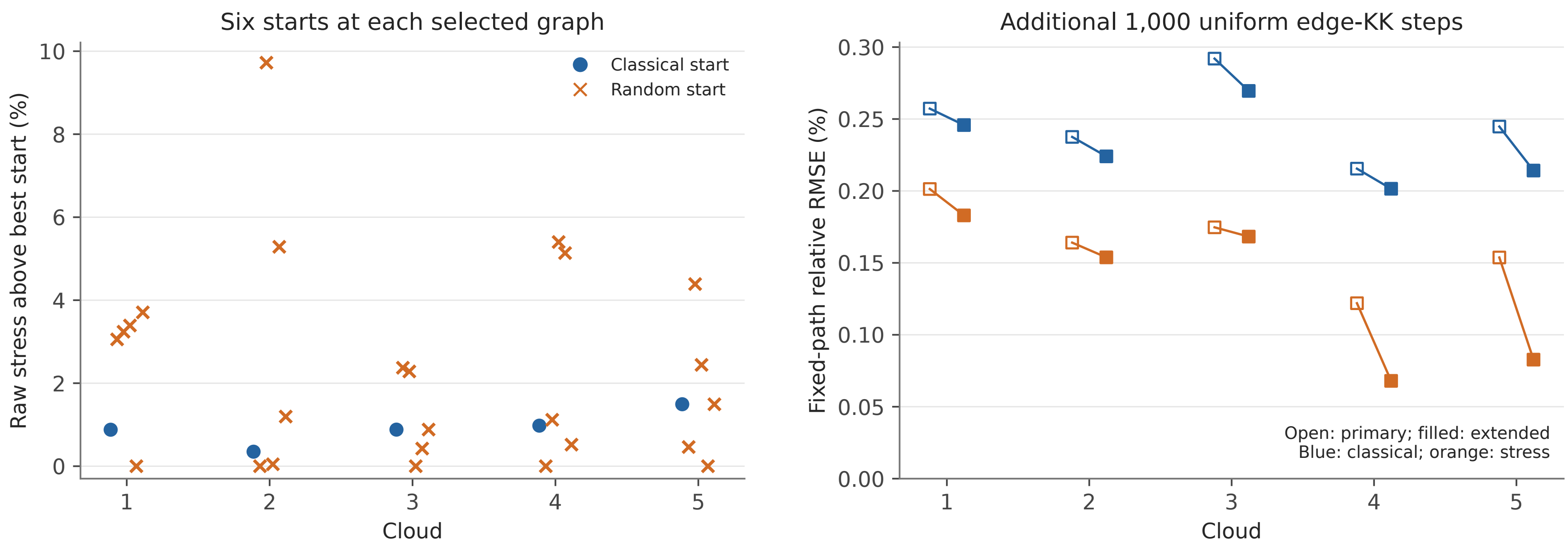
Sensitivity of geometric agreement to the graph



Similarity alignment uses known vertex correspondence. Surface RMS compares meshes over the same parameter footprint.

Both rows use similarity alignment with known correspondence. The surface statistic evaluates the same parameter footprint and mesh connectivity for every candidate. It is an agreement diagnostic, not an injectivity or topology test. Changes in the selected local MDS solution can contribute to irregular behavior across k; the figure does not isolate graph effects from optimizer basin changes.

Optimizer and budget sensitivity



At selected graphs, the worst of six achieved raw stresses exceeds the best by 2.37–9.72%. Across the full sweep, 28/150 primary starts reach the iteration limit; 0/25 selected starts do so. The other primary starts stop at the backend stress-change tolerance. The selected-graph 2,000-step tighter-tolerance continuation lowers raw stress by 0.040–0.397%. All 260 recorded edge-KK stages reach their iteration caps. The achieved finite-budget fits must not be called global optima.

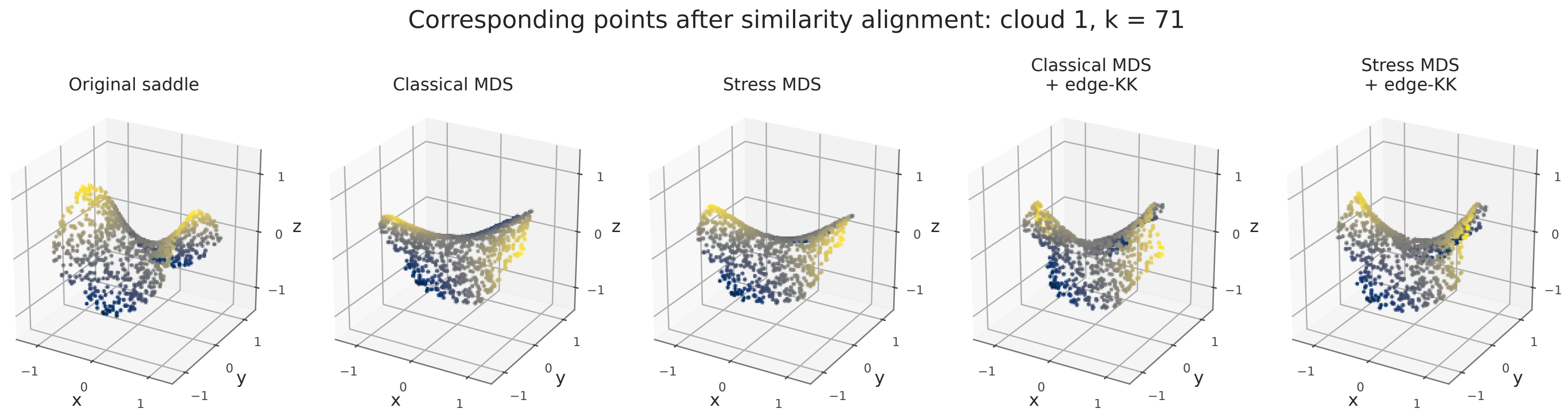
Elapsed component times were measured with concurrent cloud workers. They are not a controlled performance benchmark.

Interpretation for the software manuscript

The graph-sensitivity result belongs in the paper because it tests a practical choice that the current workflow asks the user to make. A concise main-text comparison should name classical scaling and raw-stress MDS explicitly, report both the path benefit and chord tradeoff, and show that graph/reference fidelity is a separate requirement. The complete starts, graph diagnostics, coordinate and surface checks, and additional-budget fits fit naturally in supplementary reproducibility material.

These experiments do not establish recovery of the generating saddle, a planar or tree limit, or robustness under increasing surface radius or sample size. They also do not determine a universally best k. The wider radius study remains a separate phase. The current manuscript and its original scientific caches have not been replaced by this experiment.

A fixed visual example



Identical limits and equal Euclidean units on all axes; color identifies original height. Similarity alignment cannot correct folds or correspondence error.

Cloud 1 is shown by a fixed first-cloud rule. All panels use the same limits and equal Euclidean units on the three axes after similarity alignment. Color tracks the original height of the same points. No anisotropic display rescaling is used. The quantitative comparisons across all clouds are the basis for conclusions; this single picture is an illustration.

Reproducibility

The adjacent `README.md` defines the full protocol and commands. `summary/` contains graph selection, all primary scores, 150 start records, additional-budget scores, coordinate/surface diagnostics, timings, optimizer status, source/input hashes, and validation. `coordinates.csv.gz` supports figure reconstruction without optimization. Full fit and path checkpoints live in the manuscript build tree; regenerating figures and this report uses only tracked summaries. No computation depends on private agent notes.

References

Patrick Mair, Patrick J. F. Groenen, and Jan de Leeuw. More on multidimensional scaling and unfolding in R: smacof version 2. *Journal of Statistical Software*, 102(10):1–47, 2022. doi: 10.18637/jss.v102.i10. URL https://www.jstatsoft.org/article/view/v102i10.